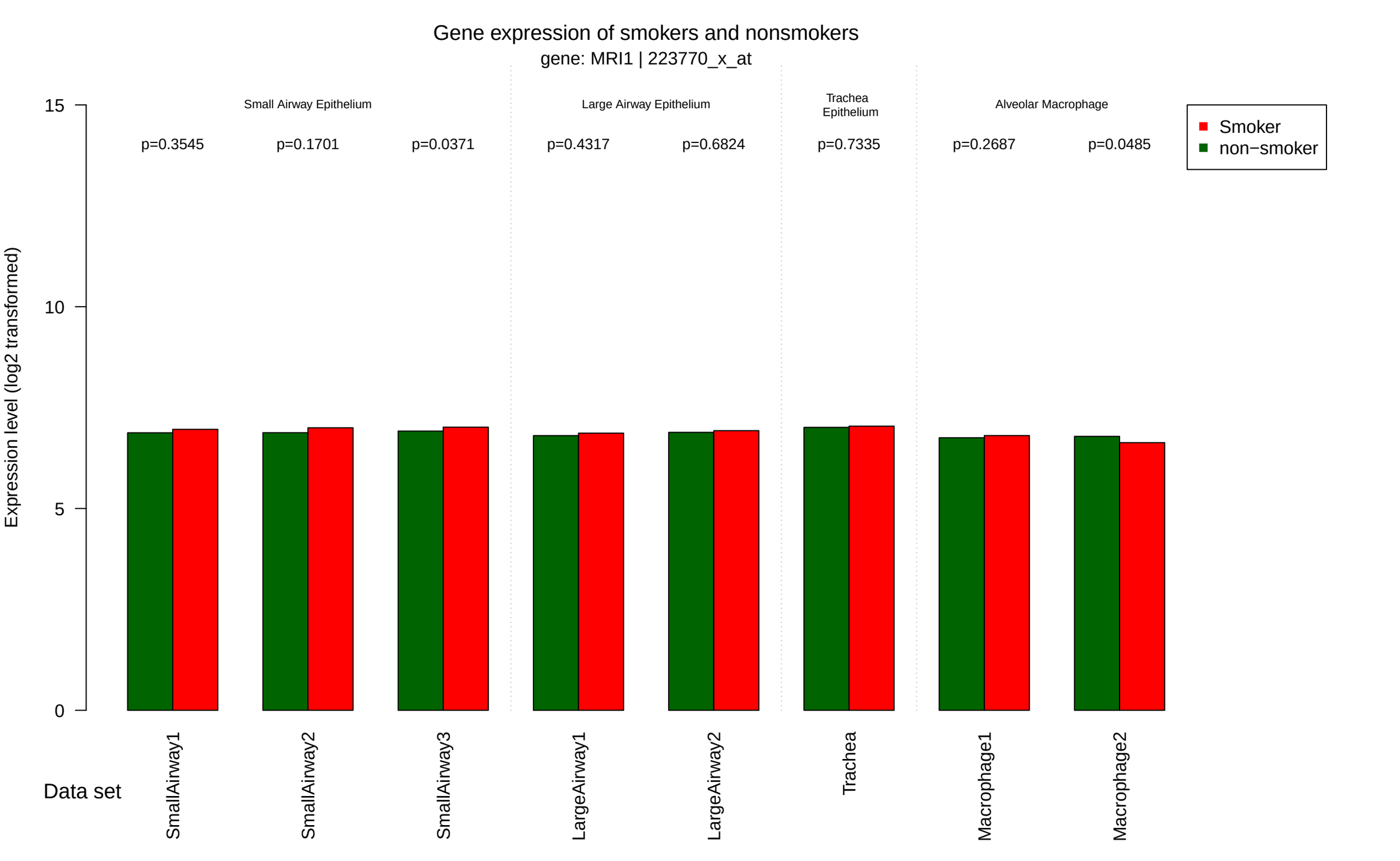
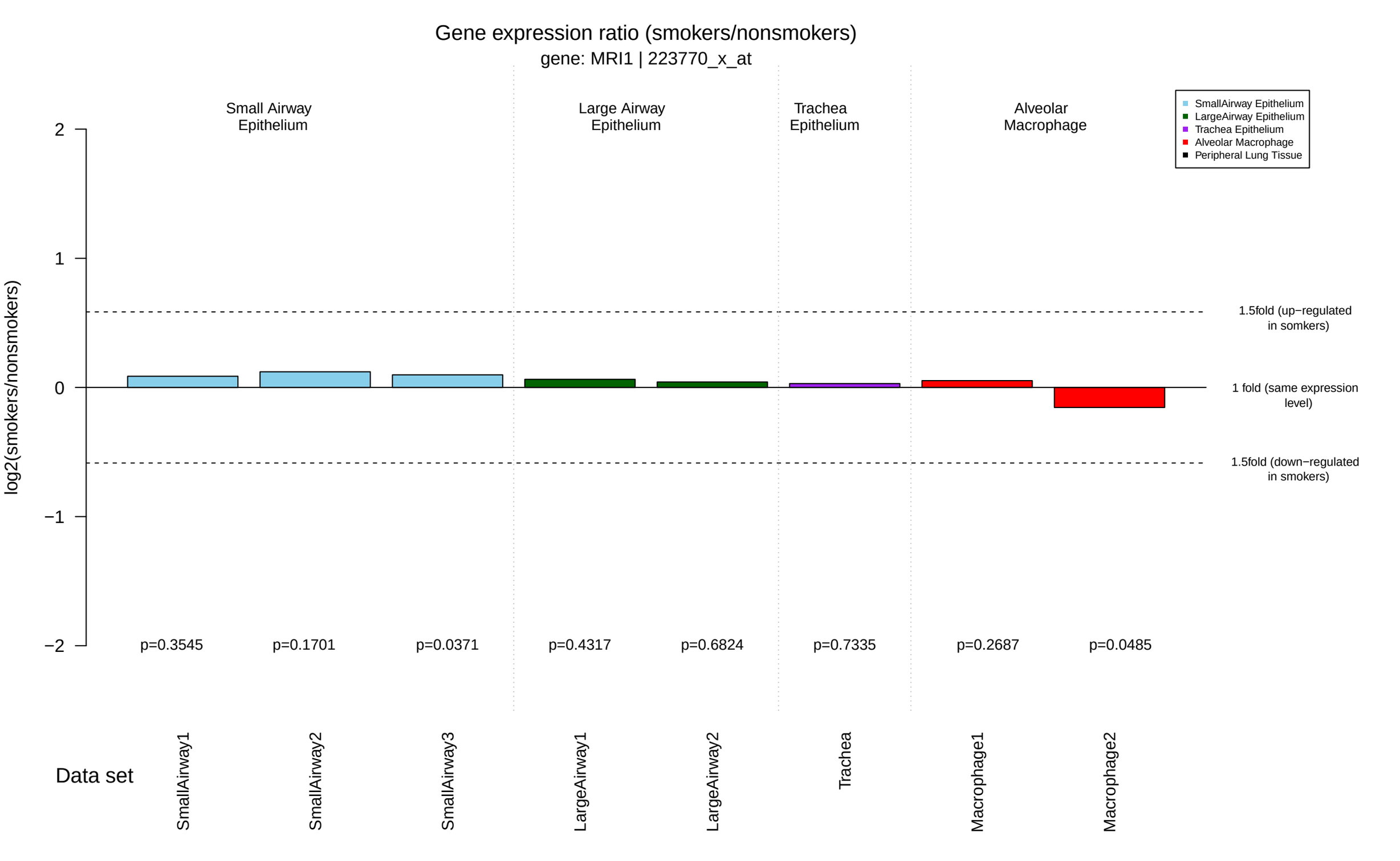
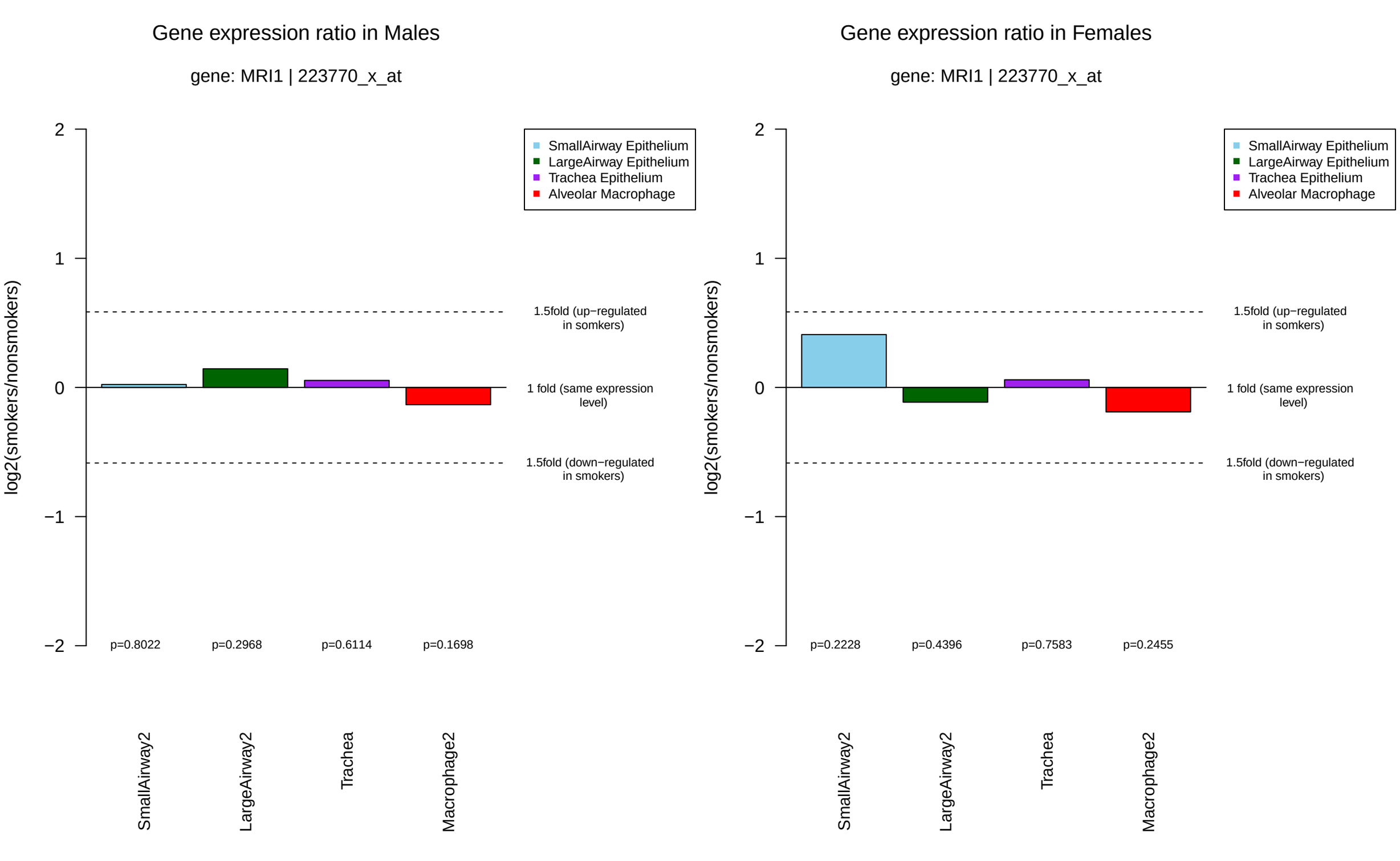
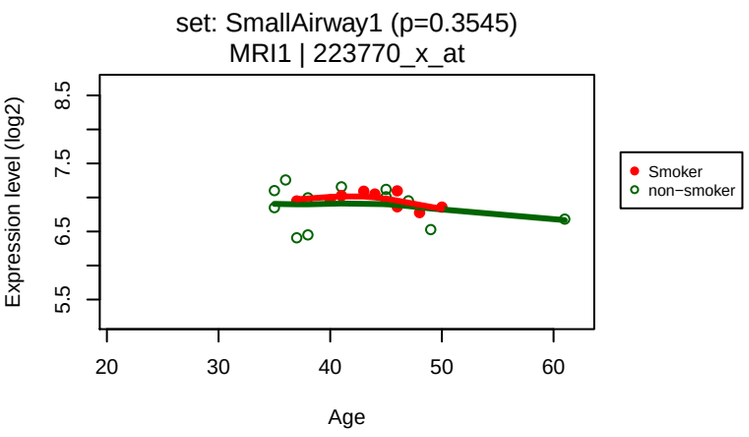
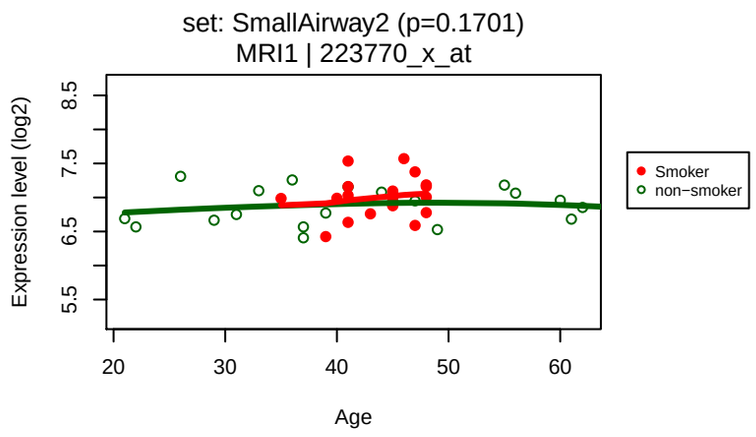
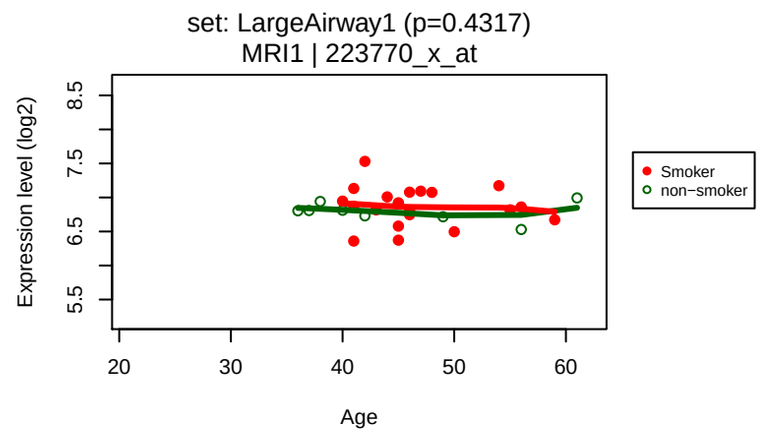

Gene expression of smokers and nonsmokers
gene: MRI1 | 223770_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3545
p=0.1701
p=0.0371
p=0.4317
p=0.6824
p=0.7335
p=0.2687
p=0.0485
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

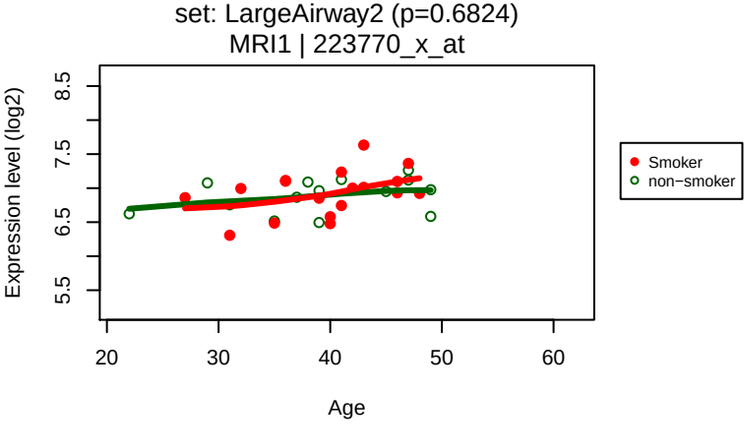

Gene expression ratio (smokers/nonsmokers)
gene: MRI1 | 223770_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3545
p=0.1701
p=0.0371
p=0.4317
p=0.6824
p=0.7335
p=0.2687
p=0.0485
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

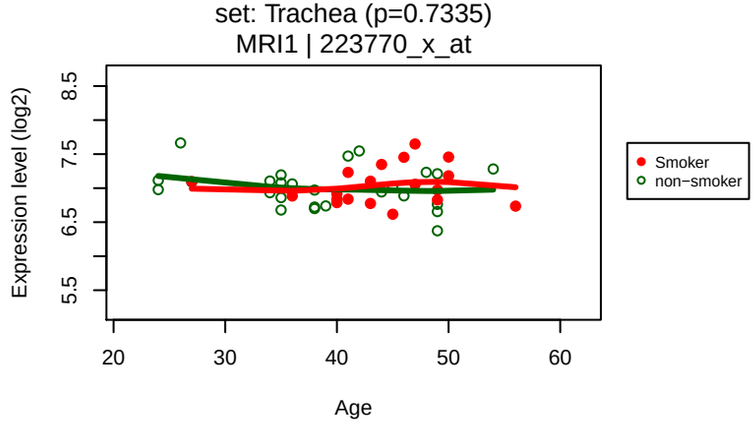

Gene expression ratio in Males
gene: MRI1 | 223770_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8022
p=0.2968
p=0.6114
p=0.1698
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRI1 | 223770_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2228
p=0.4396
p=0.7583
p=0.2455
SmallAirway2
LargeAirway2
Trachea
Macrophage2

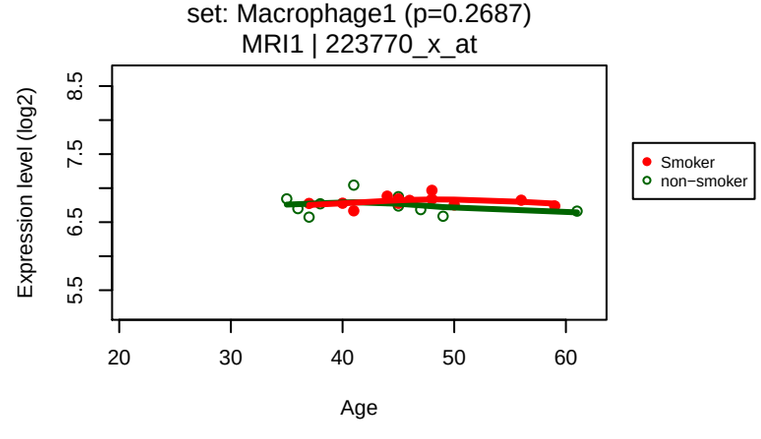

set: SmallAirway1 (p=0.3545)
MRI1 | 223770_x_at
Expression level (log2)
Age
Smoker
non−smoker

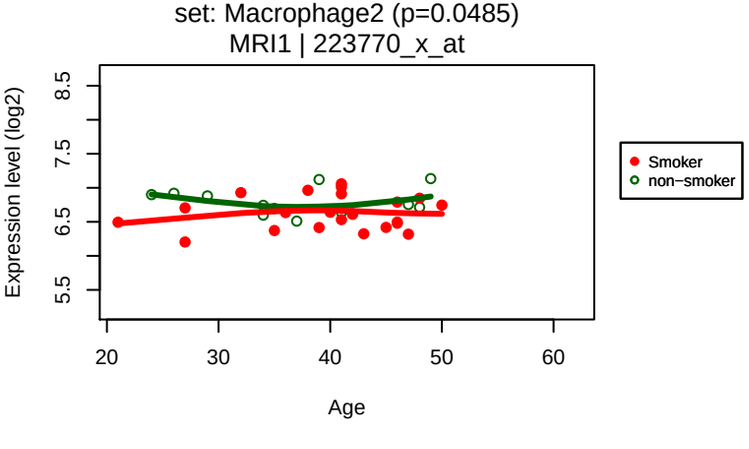

set: SmallAirway2 (p=0.1701)
MRI1 | 223770_x_at
Expression level (log2)
Age
Smoker
non−smoker

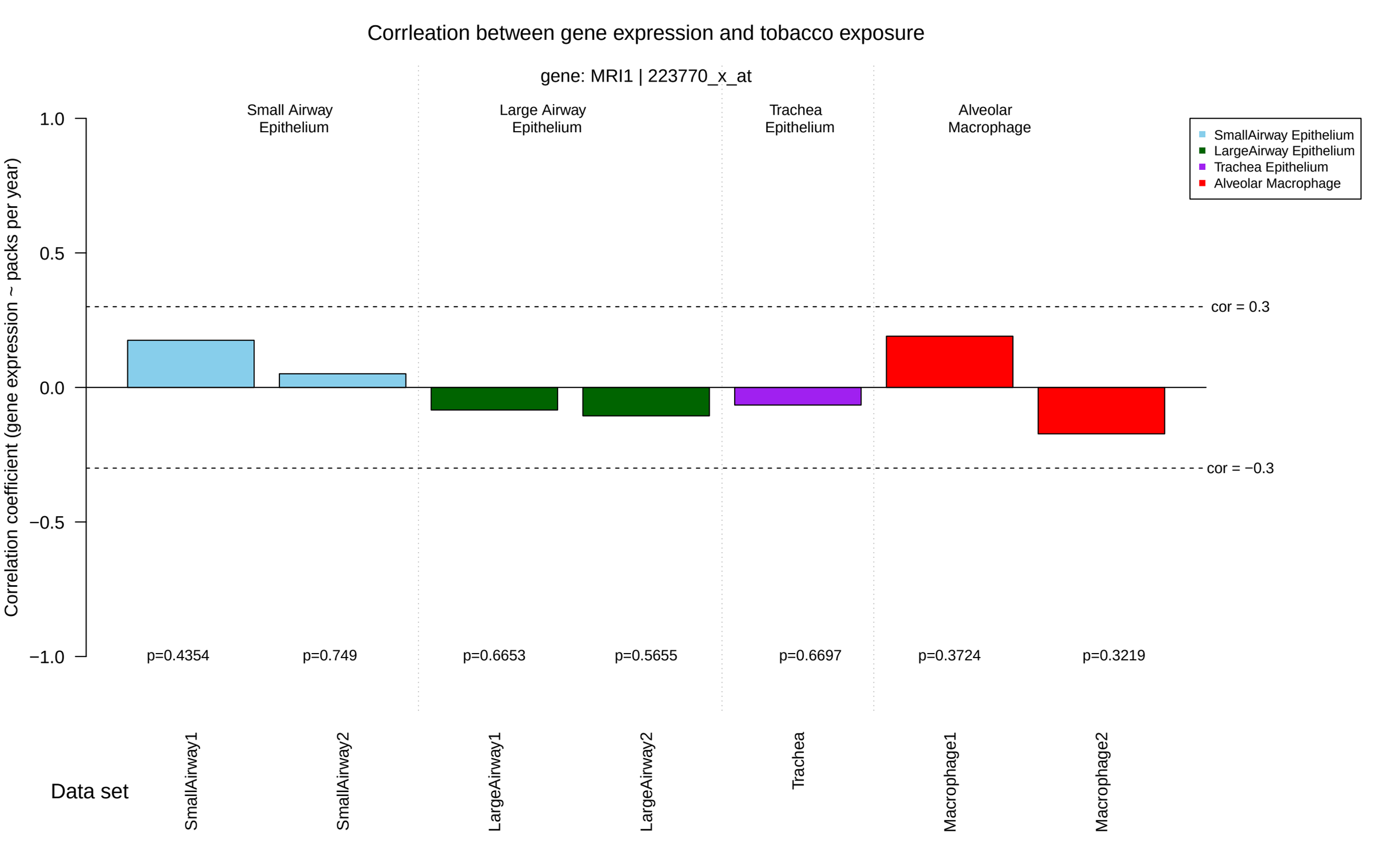

set: LargeAirway1 (p=0.4317)
MRI1 | 223770_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6824)
MRI1 | 223770_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7335)
MRI1 | 223770_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2687)
MRI1 | 223770_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0485)
MRI1 | 223770_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRI1 | 223770_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4354
p=0.749
p=0.6653
p=0.5655
p=0.6697
p=0.3724
p=0.3219
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2